

LAB 07

Title: Evaluate Prefix Expressions Using C++ with Debugging Techniques.

Objective

To learn how to debug a C++ program using Breakpoints and the Watch Window, and to understand how a prefix (Polish Notation) expression is evaluated using a stack by scanning the expression from right to left.

Procedure

1. Complete environment setup.
2. Write the Prefix Expression Evaluation code.
3. Set break points.

Breakpoint Location	Reason
for(int i = exp.length() - 1; i >= 0; i--)	To observe each character of the prefix expression being processed from right to left.
if(isdigit(ch))	To check whether the current character is a digit and should be pushed onto the stack.
else	To detect an operator and prepare to evaluate the top two elements of the stack.

4. Execute with Next Line.
5. Track the changes on Watch.

Code with break-points

```
Prefix.cpp
1  #include <bits/stdc++.h>
2  using namespace std;
3
4  int main()
5  {
6      string exp;
7      cin >> exp;
8
9      stack<int> st;
10
11     // Prefix: Right to Left
12     for(int i = exp.length() - 1; i >= 0; i--)
13     {
14         char ch = exp[i];
15
16         if(isdigit(ch))
17         {
18             st.push(ch - '0');
19         }
20         else
21         {
22             int a = st.top();
23             st.pop();
24
25             int b = st.top();
```

```

26     st.pop();
27
28     int result;
29
30     if(ch == '+')
31         result = a + b;
32     else if(ch == '-')
33         result = a - b;
34     else if(ch == '*')
35         result = a * b;
36     else if(ch == '/')
37         result = a / b;
38
39     st.push(result);
40 }
41 }
42
43 cout << "Result = " << st.top();
44
45 return 0;
46 }

```

Output

The screenshot displays a C++ IDE with the following components:

- Left Pane (Debug Console):** Shows the state of variables during execution:
 - ch = 42 '+'
 - a = 8
 - b = 2
 - result = 16
- Main Editor (Prefix.cpp):** Contains the source code for the prefix evaluator. The following lines are highlighted in orange:


```

12 for(int i = exp.length() -
13 {
14     char ch = exp[i];
15     if(isdigit(ch))
16     {
17         st.push(ch - '0');
18     }
19     else

```
- Right Pane (Output):** Displays the output of the program:


```

**532
Result = 16

```
- Bottom Pane (Debug Tools):** Includes tabs for Compiler, Resources, Compile Log, Debug (selected), and Find Results. Below these are buttons for:
 - Debug
 - Add watch
 - Next line
 - Continue
 - Next
 - Stop Execution
 - View CPU window
 - Into function
 - Skip function
 - Into